Recurrent Residual Dual Attention Network for Airborne Laser Scanning Point Cloud Semantic Segmentation

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***Abstract*— Kernel point convolution (KPConv) can effectively represent the point features of point cloud data. However, KPConv-based methods just consider the local information of each point, which is very difficult to characterize the intrinsic properties of airborne laser scanning (ALS) point clouds for complex laser scanning conditions. Therefore, we rethink KPConv and propose a recurrent residual dual attention network (RRDAN) based on the encoder–decoder structure for the semantic segmentation of ALS point cloud data. In the encoder stage, we design an attention KPConv (AKPConv) block by using a scaling factor of batch normalization to highlight the significant channel information. Then, we use the AKPConv block to develop a recurrent residual kernel attention (RRKA) module to iteratively aggregate the local neighborhood features. In the decoder stage, we design a global and local channel attention (GLCA) module with global connection and local 1-D convolution to interact the global and local information after fusing the upsampled high-level representations and the skip-connected low-level features. In addition, to reduce the influence of the long-tailed distribution of reflection intensity, we apply gamma transformation to correct the data as a normal distribution. The proposed RRDAN can achieve diversified feature aggregation to implement the refined semantic segmentation of ALS point clouds. We evaluate our method on three ALS datasets (i.e., ISPRS, DCF2019, and LASDU) to demonstrate its performance compared to a few advanced methods. The code is available at https://github.com/SC-shendazt/RRDAN.**

***Index Terms*— Attention mechanism, encoder–decoder structure, kernel point convolution (KPConv), point cloud semantic segmentation, recurrent residual structure.**

I. Introduction

With the rapid development of sensors, communication, and information processing technologies, Earth observation technology has been widely used to monitor the state of Earth by capturing multiresolution and multitype observation data in various ways [1], [2], [3]. Light detection and ranging (LiDAR) is one of the most promising technologies, which can capture the 3-D spatial data of Earth [4], [5], [6]. Unlike traditional image data, airborne laser scanning (ALS) data are stored in a point cloud format, which provides accurate spatial location information and features of ground objects. ALS point cloud has been widely applied in the fields of geohazard monitoring, high-precision mapping, large-scale urban 3-D modeling, and natural resource surveys [7], [8], [9]. In these applications, a foundation problem is to discriminate the class information of ALS point cloud data, which can be considered to recognize the class label of each point.

Semantic segmentation is a common manner to obtain the semantic class of each pixel in a traditional image, which shows a very similar process to classify each point of LiDAR [10], [11], [12]. Therefore, semantic segmentation has been successfully introduced into the field of point cloud data processing. Compared with the semantic segmentation of traditional images, point cloud semantic segmentation faces greater challenges due to its unstructured properties, limited data resolution, high sensor noise, inhomogeneous density distribution, and complex scene [13]. Recently, with the popularity of deep learning methods, significant performance has been achieved in various applications, including scene classification, target detection, and hyperspectral image classification [14], [15], [16]. Keeping up with this trend, deep learning is introduced into the field of image semantic segmentation, while it also shows competitive performance in point cloud semantic segmentation [17]. For example, to make full use of the superior performance of convolutional neural networks (CNNs), Winiwarter et al. [18] proposed to represent 3-D point cloud data by projecting them as a regular bird's-eye image and then use traditional 2-D CNNs for classification. Nevertheless, the segmentation performance is limited by the loss of information due to the 3-D-to-2-D conversion process. To preserve the spatial distribution properties of the point cloud and avoid the loss of elevation information, the voxel-based approach offers an alternative option, which aims to find a way to manage the point cloud data with certain rules for applications with 3-D CNN, including attentive feature fusion network (AF2-S3Net) [19] and fast point transformer (FPT) [20].

Manuscript received 29 March 2023; revised 6 May 2023 and 27 May 2023; accepted 8 June 2023. Date of publication 12 June 2023; date of current version 27 June 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62071340 and Grant 62201109, in part by the Natural Science Foundation of Chongqing under Grant CSTB2022NSCQ-MSX0452, and in part by the Fundamental Research Funds for the Central Universities under Grant 2023CDJXY-039. *(Corresponding author: Fulin Luo.)*

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Digital Object Identifier 10.1109/TGRS.2023.3285207

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The above projection- and voxel-based methods both require the conversion of raw point cloud data into an intermediate format for the standard convolution operation, which will destroy the geometric structure of point clouds. Therefore, point-based methods have been used to directly explore the intrinsic relationships of point cloud data, including pointwise multilayer perceptron (MLP), graph-based, and convolution-based [21], [22]. PointNet [17], [23] is a milestone architecture in the pointwise MLP family, which uses the shared MLP to directly learn each raw point cloud feature and eventually aggregate global information with maxpooling function. Nevertheless, the efficiency of PointNet to capture local features of point clouds is relatively low, resulting in poor robustness to complex scenarios. In response to this challenge, many improvement methods based on pointwise MLP have been proposed, such as PointNet++ [24], RandLA-Net [25], VD-LAB [26], and RFFS-Net [27], which can achieve good performance in local–global feature representation. However, the pointwise MLP-based methods pay little attention to the underlying geometric information of each point. To better capture the local geometric relationships in high-dimensional space, a series of graph-based algorithms is designed to characterize the topologic structure between a point and its neighbors, including DGCNN [28] and GACNet [29]. Different from the pointwise MLP approaches, graph-based methods provide explicit representations of intrinsic geometric properties for point clouds in the Euclidean space. Noticeably, for the abovementioned pointwise MLP and graph-based methods, they typically rely on standard convolution to explore the information of point cloud data with a fixed convolutional region, which is extremely sensitive to noise, affecting the accuracy of segmentation and decreasing the discriminative performance of features.

To adaptively explore the spatial relationship of each point, a kernel point convolution (KPConv) [30], [31] was designed to characterize 3-D point clouds by considering the Euclidean distance of a center point to each point with different weights. Based on this framework, a lot of related methods were introduced for point cloud segmentation, such as MappingConv [32], DenseKPNET [33], and PAConv [34]. To better obtain the neighborhood information of point clouds, a lightweight position pooling (PosPool) [35] operator without weight learning is proposed to aggregate local features. To precisely capture the diverse relations between points from different semantic parts, an anisotropic adaptive KPConv (AdaptConv) network [36] is presented that can dynamically learn the features of each point, which represents the diversity of kernels to each pair of points rather than relying on the predefined weights. To establish long-range contextual relationships, a transformer structure [37] was introduced into the KPConv-based network, which adaptively captures global information with transformer. To explore contextual information at the object and point level for ALS point clouds, a local and global encoder network (LGENet) [13] with KPConv was proposed to learn more representative features from ALS data.

The original KPConv network just considers the single information of each point in a certain local region, which neglects the underlying diversity information of each point. However, the ALS point clouds have a complex distribution of land covers. In the downsampling stage, the KPConv network pays less attention to the fine-grained local neighborhood features, while it lacks information interactions within layers in the upsampling stage. Therefore, the original KPConv network is very difficult to comprehensively and effectively represent the intrinsic properties of ALS point clouds. In this article, we rethink KPConv and propose a recurrent residual dual attention network (RRDAN) based on decoder–encoder structure, as shown in Fig. 1.

The proposed RRDAN consists of an attention KPConv (AKPConv) block, a recurrent residual kernel attention module (RRKA), and a global and local channel attention (GLCA) module. AKPConv enhances the local feature representation of KPConv via the attention of batch normalization weights. We utilize the recurrent residual operation on AKPConv to construct the RRKA module for aggregating the diversified point features, which represents the local neighborhood information of point clouds with feature accumulation learning. In the upsampling stage, GLCA performs the interaction of global and local features, and the geometric scale of point cloud data is restored layer by layer. Unlike the previous schemes in the literature [13], the RRDAN network captures the geometric neighborhood features of ALS point clouds in a comprehensive and effective manner, and the global information is better represented due to the advantage of feature accumulation and attention. The main contributions of this article are summarized as follows.

1) To enhance the representation of significant features, we design an AKPConv block to weight the channel information by using the batch normalized scale factor, which improves the significant information representation and reduces the influence of outlier features.
2) Based on the AKPConv block, we develop an RRKA module through the iteration cumulation of local features to aggregate the features of neighbor points. The RRKA module can effectively extract feature information from different neighborhood point cloud data.
3) We design a GLCA module to combine the global and local properties, which facilitates the interaction of global and local information between shallow and deep features to enhance the comprehensive feature representation for the improvement of segmentation performance.
4) A feature accumulation and interaction framework for point cloud information extraction, RRDAN, is proposed based on AKPConv, RRKA, and GLCA. RRDAN extracts the diversified discriminating features and shows the powerful ability to point cloud segmentation with the comprehensive experiments on three datasets.

The remainder of this article is organized as follows. In Section II, the details of RRDAN are depicted, including AKPConv, RRKA, GLCA, and reflection intensity transformation. Section III describes the datasets, evaluation metrics, and experimental results. Section IV concludes this article.

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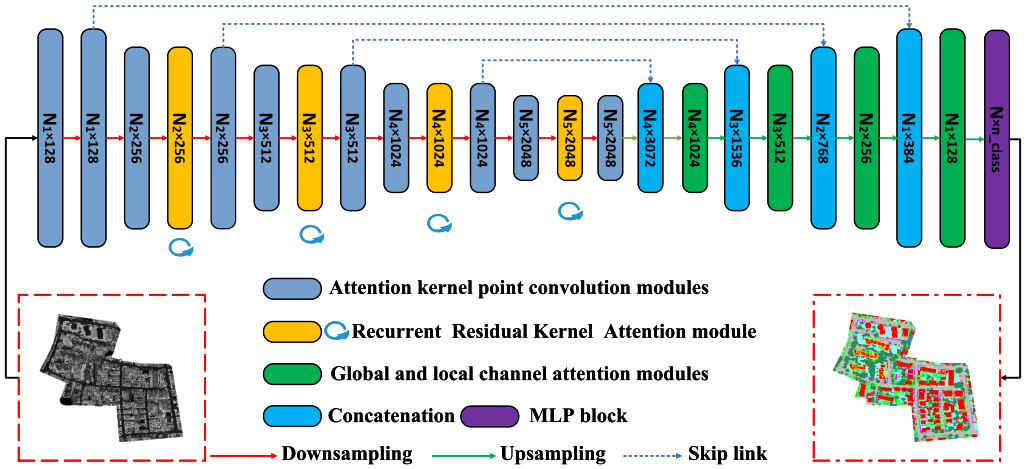


Fig. 1. Detailed network architecture.

II. Methodology

A. Overall Architecture

The proposed RRDAN is based on KPConv and encoder–decoder structure. Fig. 1 shows the detailed architecture of RRDAN, which is implemented by a classic encoder–decoder architecture with skip connections. We construct an AKPConv module to enhance the significant local features. Based on AKPConv and recurrent operation, we develop an RRKA to aggregate the local information in the downsampling stage. As shown in Fig. 1, the sampled number of point clouds is set to $N_1 > N_2 > N_3 > N_4 > N_5$.

The decoder has four layers after the above encoding layers. For each layer in the decoder, we first use the nearest upsampling algorithm to obtain the pointwise features. Next, the upsampled features are concatenated with the shallow features from encoding layers through skip connections. Then, a GLCA module is designed to implement the interaction of local and global information between the shallow and deep features. At the end of the network, one MLP is applied to predict the semantic label for each point.

B. Attention KPConv

KPConv is a 3-D kernel convolution for flexible feature representation in a spherical 3-D space. It has two forms, including rigid and deformable. Thomas et al. [30] showed that rigid convolutions perform better than deformable convolutions in different scenes, so we use rigid convolutions in our experiments. If not specified, KPConv means rigid KPConv in the following.

In the previous research, many works tried to improve the performance of networks by suppressing unimportant features through the attention mechanism. For instance, Hu et al. [38] achieved adaptive learning on the features of the channels by modeling the interdependence between the channels. However, these previous attention modules could not be integrated inside the KPConv model. Instead, they are simply added to the network, ignoring the importance of features during the KPConv operation. As ALS point cloud data are acquired by airborne LiDAR devices from a top-view perspective, which presents occlusions and extreme scale variations for most objects. According to [30], to ensure the sphere sampling based on radius neighborhood without empty set, the grid radius needs to be large enough to cover the whole ALS point cloud scene. Furthermore, the performance of KPConv with large radius sampling is poor. Therefore, an efficient AKPConv module is designed with channel feature weighting based on the scale factor of batch normalization to reduce the influence of outlier features and enhance the feature representation ability for ALS point cloud data with large grid radius sampling. In addition, the data shape of each layer is $N_i \times F$, where $i = \{1, 2, 3, 4, 5\}$ and F is represented as the feature channel and is set to 128, 256, 512, 1024, and 2048 in the downsampling stage.

For the sake of clarity, a point x_i and its features f_i are from the point set $P \in \mathbb{R}^{N\times 3}$ and the corresponding feature set $F \in \mathbb{R}^{N\times D}$. For KPConv, the point convolution of $\mathcal{F}$ with a kernel function κ at a point $x \in \mathbb{R}^{N\times 3}$ is defined as

$$(\mathcal{F} * \kappa)(x) = \sum_{x_i \in \mathcal{N}_x} \kappa(x_i - x) f_i \tag{1}$$

where $\mathcal{N}_x$ is a set of neighbors around x within a fixed radius $r \in \mathbb{R}$ and $\mathcal{N}_x = \{x_i \in P \mid ||x_i - x|| \leqslant r\}$. x_i is a neighbor of x and its corresponding features are $f_i \in \mathbb{R}^{N_i \times D_i}$, where D_i is the channel number of features.

For simplicity, we set the input of kernel function κ as $\ell_i = x_i - x$, and $\{\ell_i \in \mathbb{R}^3 \mid ||\ell_i|| \leq r, i \in 1, 2, \ldots, N'\} \subset S_r^3$, where ℓ_i is the relative position of neighboring points to the central point x, N' is the number of neighbors, and S_r^3 represents the spherical domain. Similar to the traditional convolution kernel, the kernel function κ applies different weights to different regions in S_r^3, and different regions denote a series of kernel points. Let $\{\tilde{x}_k \in \mathbb{R}^3 \mid k < K\} \subset \mathcal{S}_r^3$, where $\tilde{x}_k$ is a 3-D position in S_r^3 and n_k is the number of kernel points for the kernel function κ. The weight matrix of the corresponding kernel point is represented as $\{W_k \mid k < n_k\} \subset \mathbb{R}^{D_{\text{in}} \times D_{\text{out}}}$, mapping features from dimension D_{in} into D_{out}. The kernel function κ for any input $\ell_i \in S_r^3$ is defined as

$$\kappa(\ell_i) = \sum_{k<n_k} h\left(\ell_i, \tilde{x}_k\right) W_k \tag{2}$$

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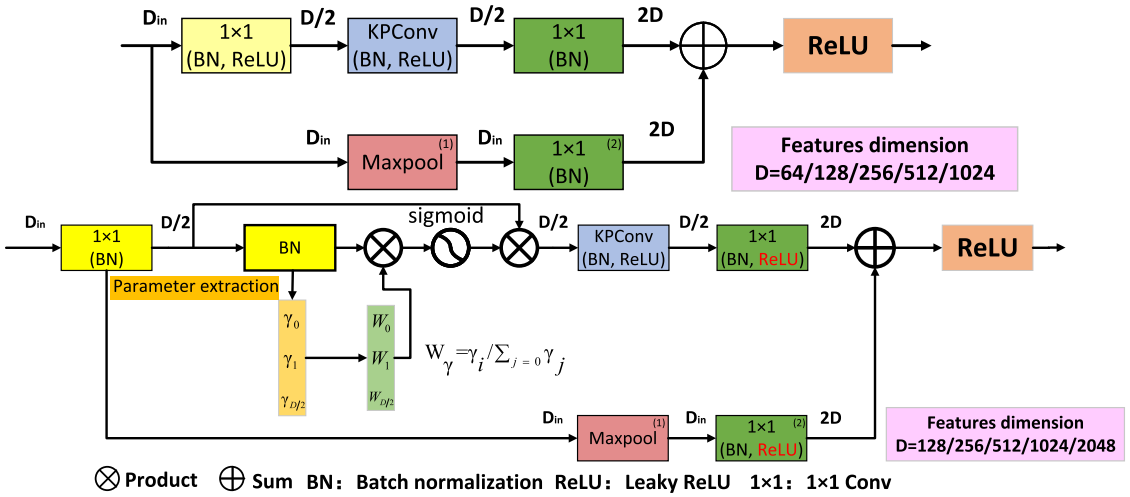


Fig. 2. (Top) KPConv block used in [30]. (Bottom) AKPConv block proposed in this article. For skip connections, if $D_{\text{in}} = 2D$, the maxpooling (1) and the 1×1 convolution (2) will be simultaneously used or it only uses the 1×1 convolution.

where h is the linear correlation between $\tilde{x}_k$ and ℓ_i, whose value should be higher when $\tilde{x}_k$ is closer to ℓ_i. The correlation function h is defined as

$$h\left(\ell_i, \tilde{x}_k\right) = \max\left(0, 1 - \frac{\|\ell_i - \tilde{x}_k\|}{\sigma}\right) \tag{3}$$

where σ is a parameter to control the influence of kernel point distance, which is set based on the input density.

To enhance the feature representation of KPConv, channel attention is constructed to highlight the key features. For high-dimensional features, there are dependencies between channels, which can improve feature identifiability for different semantic categories. According to [39], the scale factor γ of batch normalization can represent the dependencies between channels. Therefore, we adopt the scale factor to construct channel attention for KPConv. Let f_{in} and f_{out} be the input and output of a batch normalization layer, respectively. For the current batch $\mathcal{B}$, the batch normalization layer performs the following transformation:

$$f_{\text{out}} = \gamma \frac{f_{\text{in}} - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \varepsilon}} + \beta \tag{4}$$

where $\mu_{\mathcal{B}}$ and $\sigma_{\mathcal{B}}$ are the mean and standard deviation of a batch, respectively, and γ and β are trainable affine transformation parameters. γ is the scale factor related to channels, and the attention weights are obtained as $W_\gamma = \gamma_i / \sum_{j=0} \gamma_j$. The attention output f_c can be represented as

$$f_c = \text{sigmoid}\left(W_\gamma \cdot f_{\text{out}}\right) \cdot f_{\text{in}}. \tag{5}$$

Based on the attention features, they are input into the point convolution as follows:

$$F_1 = \text{Conv}_{1\times1}(\text{KPConv}(f_c)). \tag{6}$$

To preserve the input features, we add the skip connection into the AKPConv block. For the skip connection, the maxpooling is an optional operation, which will be used when $D_{\text{in}} = 2D$. The operation can be denoted as

$$F_2 = \begin{cases} \text{Conv}_{1\times1}(\text{Maxpool}(f_{\text{in}})), & D_{\text{in}} = 2D \\ \text{Conv}_{1\times1}(f_{\text{in}}), & \text{otherwise.} \end{cases} \tag{7}$$

The output of AKPConv can be represented as

$$F_{\text{AKPConv}} = \text{ReLU}(F_1 + F_2). \tag{8}$$

C. Recurrent Residual Kernel Attention

The complexity of ALS point cloud data presents a significant challenge for feature extraction. In the case of AKPConv block, it only considers the individual information of each point within a local region, which proves to be a difficult task for characterizing the intrinsic properties of ALS point clouds, especially for complex laser scanning conditions. To address this issue, we propose the RRKA module based on AKPConv. The RRKA module effectively captures the complex and diverse features from ALS point cloud data by iteratively accumulating local features. Moreover, the RRKA module captures broader features and learns more refined representations of ALS data, which enhances the ability of networks to extract meaningful features. The overall structure is shown in Fig. 3. The overall module consists of several AKPConv layers, recurrent blocks, and MLP blocks. The main goal of this module is to aggregate the local features and generate the diversified features.

In this module, based on AKPConv, the recurrent point convolution (RPConv) block is designed to learn the cumulative neighbor features of point clouds, which considers the current layer information and its previous recurrent information. Let x_r as the input sample of RPConv with T time recursions at the lth layer, and the output at the tth time recursion can be expressed as follows:

$$x_r^t = \text{AKPConv}\left(x_r^f(t)\right) + \text{AKPConv}\left(x_r^r(t-1)\right) \tag{9}$$

where $t = 1, 2, \ldots, T$, AKPConv$(\cdot)$ is the AKPConv operation, x_r^t is the output of RPConv operation at the tth time recursion, $x_r^f(t)$ denotes the input x_r of RPConv at the tth time, and $x_r^r(t-1)$ is the output of the $(t-1)$th time recursion of x_r. When $t = 0$, (10) only uses the AKPConv operation to the input x_r.

Due to adding the recurrent operation, the computational cost naturally increases. To satisfy the computational efficiency

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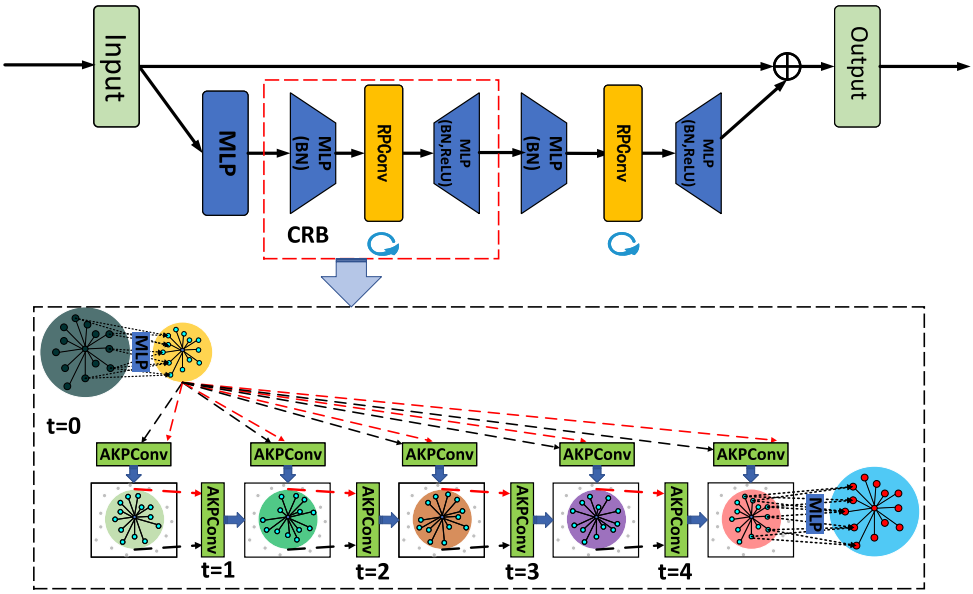


Fig. 3. Structure of the RRKA module, which consists of MLP and AKPConv with residual information and recurrent operation.

of RPConv, we first adopt MLP to compress the dimensionality of features before using RPConv and then use MLP to recover the dimensionality of features. Therefore, we can construct a compressed RPConv block (CRB) based on RPConv and MLP to improve the computational efficiency. For the input x_{cin}, the output x_{cout} of CRB can be used as

$$x_{\text{cout}} = \text{ReLU}(\text{BN}(W_2 * \text{RPConv}(\text{BN}(W_1 * x_{\text{cin}})))) \quad (10)$$

where W_1 and W_2 are the learnable parameters for MLP and PRConv$(\cdot)$ is the RPConv operation.

Then, based on CRB, we develop the RRKA module with the residual information that can improve the network trainability and the reuse capability of features. This proposed module effectively works on the recurrent operation to aggregate the local features and enhance the diversity representation of point cloud features. For the input x_{in} of the RRKA module, the output of RRKA can be obtained by

$$x_{\text{out}} = \text{CRB}(\text{CRB}(W_0 * x_{\text{in}})) + x_{\text{in}} \quad (11)$$

where W_0 is the learnable parameter of MLP and CRB$(\cdot)$ is the CRB operation.

D. Global and Local Channel Attention

The previous KPConv-based networks can represent the geometric distribution of 3-D point cloud data well. However, these networks only use MLP to restore the scales in the upsampling state, which just reveals the global properties between the shallow and deep features. This upsampled procedure neglects the importance of local information, which lacks the feature interaction between the global information and the local information. Therefore, these networks are very difficult to comprehensively explore the intrinsic features for the complex ALS point cloud scenes in the encoder stage. We design an efficient global and local information interaction module based on attention to simultaneously represent the global and local properties of point clouds after the concatenation features

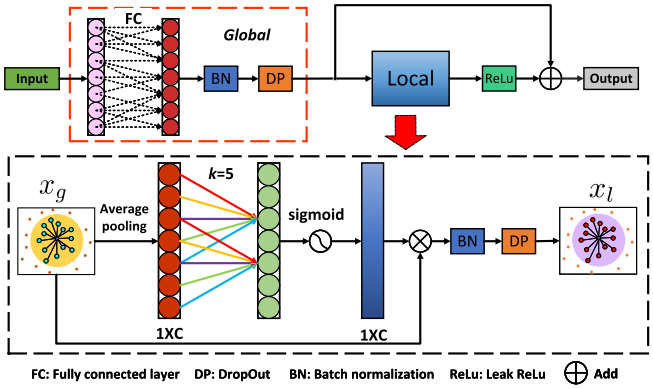


Fig. 4. Framework of the GLCA module to represent the global and local properties of point clouds.

of low-level information and high-level information, as shown in Fig. 4.

In this module, we adopt the full connection manner to implement the channel information fusion of low- and high-level features under the global space. Then, local attention is designed to highlight the significant local information, which uses the 1-D convolution of the average-pooling features of point cloud to obtain the attention weights. The local properties can be achieved by a fixed kernel for 1-D convolution. The attention features can be generated by adding the attention weights to the point cloud features. Finally, we implement the information interaction of global fusion features and local attention features.

Let the input of the module be $x_{\text{in}} \in \mathbb{R}^{N \times C}$, where N and C are the number of upsampling point cloud and the dimensionality of features, respectively. For the global fusion, it can be denoted as

$$x_g = \text{BN}\left(W_g * x_{\text{in}}\right) \quad (12)$$

where W_g is the full connection weight to fuse the low-level information and high-level information.

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In the local feature operation, we use the average pooling and 1-D convolution to obtain the attention weights as follows:

$$\omega = \text{sigmoid}\big(W_k * \big(\text{avgpooling}\big(x_g\big)\big)\big) \tag{13}$$

where $\text{avgpooling}(x_g) = (1/N)\sum_{i=1}^{N} x_g$ is the channelwise average-pooling operation. W_k is the learnable local 1-D kernel convolution weights with a size of k (it is set to 5 in this article), which can reveal the local properties of point clouds. The kernel weights can be represented as

$$W_k = \begin{bmatrix} w^{1,1} & \cdots & w^{1,k} & 0 & 0 & \cdots & \cdots & 0 \\ 0 & w^{2,2} & \cdots & w^{2,k+1} & 0 & \cdots & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & w^{C,C-k+1} & \cdots & w^{C,C} \end{bmatrix}. \tag{14}$$

Based on the attention weights, we can achieve the local attention features in the following:

$$x_l = \text{BN}\big(\omega \odot x_g\big) \tag{15}$$

where $\odot$ denotes the elementwise multiplication.

Finally, the information interaction is designed to aggregate the global and local features, which is represented as

$$x_{\text{out}} = x_g + x_l \tag{16}$$

where x_{out} is the output of the GLCA module.

E. Gamma Transformation for Reflection Intensity

ALS not only accurately obtains the coordinate information but also records the reflection intensity value of each target point. According to existing research, the processing method of reflection intensity is complex and nonuniform in different scenes. For example, Yan and Shaker [40] proposed a subhistogram matching technique for the radiometric normalization of overlapping intensity data. The previous methods are very complex to process point clouds in real applications. Therefore, it becomes particularly important to develop a simple and efficient intensity preprocessing method for the uneven distribution of ALS point cloud data.

The intensity value of ALS point cloud can be considered as a returned discrete integer value of echo power of target through ALS after preprocessing. The echo power P_r of ALS can be obtained according to the laser echo equation [41]. After a series of signal processing operation, the ALS intensity value can be denoted as

$$I = \psi(P_r) \tag{17}$$

where ψ denotes the signal processing operation.

Inspired by the gamma transformation in image processing, we perform a gamma correction of the reflection intensity of ALS, which is a nonlinear approach to enhance the contrast ratio of the intensity features of point cloud data. The gamma transformation can be defined as follows:

$$I_\gamma = \phi \cdot \left(\frac{I}{I_{\max}}\right)^{\gamma} \tag{18}$$

where I_γ is the intensity value after gamma transformation, γ is a parameter, and $I_{\max}$ is 255. To obtain a good contrast ratio, it is very important to set a proper γ, which should be neither too small nor too large. As shown in Fig. 5, a large or small value of γ will result in the long-tailed distribution of intensity features, where we set γ to 0.5 in this article. Let $\phi = 255$ so that the reflection intensity is mapped into the rang of 0 and 255, and the point cloud intensity values can be visualized in a gray space.

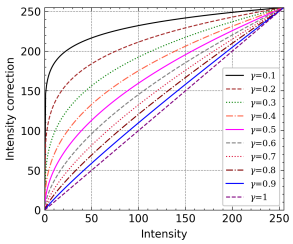


Fig. 5. Gamma transformation for the intensity of point clouds.

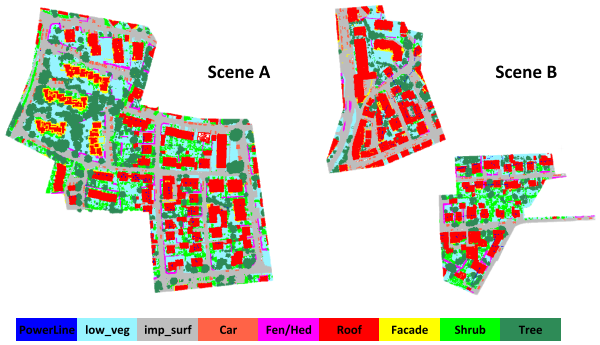


Fig. 6. ISPRS 3-D labeling dataset. (Left) Scene A is used for model training. (Right) Scene B is used for model evaluation. The point categories in each scene are presented in the colors of the legend at the bottom.

III. Experiment

A. Experimental Datasets

To evaluate the performance of the proposed method, we conduct experiments on the ISPRS dataset [42]. This dataset includes two scene areas with various object classes. This dataset was captured from the Leica ALS50 system with a field of view of 45° at an average flying height of 500 m above the ground. Nine semantic categories were labeled, including power line, low vegetation (low_veg), impervious surface (imp_surf), car, fence/hedge (fen/hed), roof, facade, shrub, and tree. In the dataset, the first scene (shown in the left of Fig. 6) with 753 876 points was used as the training set, and the other scene (shown in the right of Fig. 6) with 411 722 points was considered as the test set. The detailed category information of each scene is shown in Table I.

DFC 2019 is an ALS dataset published by the IEEE Geoscience and Remote Sensing Society (IGRSS) in 2019 [43]. The dataset was captured in two cities (Omaha, NE, and Jacksonville, FL, USA) with the pulse interval of 0.8 m, covering about 100 km^2 and consisting of 200 million points. The dataset was labeled into five semantic categories, i.e., ground, high vegetation, building, water, and bridge deck.

LASDU [44] is a large ALS point cloud dataset captured using Leica ALS70 from an altitude of around 1200 m over

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TABLE I
CATEGORY INFORMATION OF POINT CLOUD DATA FOR EACH SCENE

Categories	Scene A	Scene B
power line	546	600
low_veg	180,850	98,690
imp_surf	193,723	101,986
car	4,614	3,708
fen/hed	12,070	7,422
roof	152,045	109,048
facade	27,250	11,224
shrub	47,605	24,818
tree	135,173	54,226
Total	753,876	411,722

the Heihe River valley in northwest China. It contains about 3.12 million points categorized into ground, building, tree, low vegetation, and artifacts, covering a 1-km^2 urban area with high-density residential and industrial buildings.

B. Evaluation Metrics and Loss Function

To evaluate the segmentation results, we used the precision, recall, average F1 score (Avg.F1), and overall accuracy (OA) to quantify the segmentation performance of 3-D point cloud. In general, the OA is used to represent the overall classification results of all categories, which is defined as the percentage of correctly classified points in the total test points. Moreover, the F1 score considers the precision and recall of the classification model and is more suitable for this case where the categories are unevenly distributed. The precision, recall, and F1 score for each category are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{21}$$

$$F1 = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}} \tag{22}$$

where true positive (TP), true negative (TN), false positive (FP), and false negative (FN), respectively, denote the number of correct prediction as positive points, the number of correct prediction as negative points, the number of incorrect prediction as positive points, and the number of incorrect prediction as negative points.

Generally, the category distribution of each scene is extremely uneven, which reveals the proportion of the point number of each category. For example, on scene A in Table I, the point clouds labeled as power line only have 0.07% of the points, while the point clouds labeled as impervious surface account for 25.69%. To directly train on such an unbalanced dataset, there will be a potentially serious negative impact on the overall performance of deep network [45]. To address the problem, we introduce the focal loss function [46], and it is expressed as

$$L_{\text{FL}} = -\frac{1}{N}\sum_{j=1}^{N} \alpha_t \left(1 - \rho_{jc}\right)^{\lambda} \log\left(\rho_{jc}\right) \tag{23}$$

where λ is a parameter (generally, it is set to 2), α_t denotes the category weight parameter, N denotes the number of point clouds, and ρ_{jc} denotes the probability that the jth sample is classified into category c.

C. Experiment on the ISPRS Benchmark Dataset

1) Implementation Details: To obtain a uniform intensity distribution, we first used the gamma transformation to preprocess the intensity values. We used the coordinates (X, Y, Z) and the gamma transformation intensities as the input of the segmentation algorithm. In the training data, we used a grid sampling [30] to obtain the subsampled point cloud data, which can reduce the influence of the uneven distribution of points. The spheres for AKPConv with the radius of 20 m were randomly selected from the subsampled point cloud data. Then, the intensities and the absolute XYZ coordinates within the spheres were used as input features. For data augmentation, the input points were randomly rotated around the Z-axis to improve the network robustness to orientation. In addition, the random noise with 0.1 cm was added to the XYZ coordinates, which can enhance the network robustness to noise.

The encoder of the proposed network consists of five convolutional layers and their output feature dimensions are 128, 256, 512, 1024, and 2048. Their grid sampling intervals are 0.45, 0.9, 1.8, 3.6, and 7.2 m, and the corresponding convolution radii incrementally increase as 1.125, 2.25, 4.5, 9, and 18 m. For the RRKA module, the number of recursion t is set to 4. The decoder of the segmentation network consists of four GLCA modules, which can recover the features to the corresponding dowsampling scale. In GLCA, the local 1-D convolution kernel k is set to 5.

The training and testing process of the proposed method was conducted with the framework of Pytorch on an NVIDIA 3090 24-GB GPU. We adopted the stochastic gradient descent (SGD) optimizer to optimize the network weights. The focal loss function was used to solve the category imbalance problem, which assigns a small weight to well-categorized samples. The learning rate started from 0.001 with a decay rate of 0.8 at each epoch. The model was trained with 40 epochs to achieve the convergence and each epoch has 1000 iterations. For GLCA, it contains two dropout layers with the dropout rate of 0.2. For the testing stage, we randomly selected the spheres for each point with 50 times in the test area as the input of network to obtain a predictive probability.

2) Classification Results: To show the classification of RRDAN on the ISPRS dataset, we trained the proposed model until convergence on the training set. Then, we tested the classification results on the test set, and the confusion matrix is shown in Table II. In Table II, each row denotes the number of points predicted as a certain category.

According to Table II, the proposed model can identify most of the nine categories very well, especially for the categories of low vegetation, impervious surfaces, roof, and tree, whose

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TABLE II

CONFUSION MATRIX OF THE PROPOSED NETWORK ON THE ISPRS BENCHMARK DATASET. PRECISION (%), RECALL (%), AND F1 SCORE (%) ARE REPORTED FOR EACH CLASS. THE OA IS 84.9% AND THE AVERAGE F1 SCORE IS 74.6%

Category	power	low_veg	imp_surf	car	fence/hedge	roof	facade	shrub	tree	Precision	Recall	F1 score
power	397	0	0	0	0	114	22	7	60	79.4	66.2	72.2
low_veg	0	81025	9172	79	293	277	312	5860	1672	81.3	82.1	81.7
imp_surf	0	7845	93652	36	13	148	33	237	22	90.6	91.8	91.2
car	0	291	73	2961	105	1	5	272	0	89.9	79.9	84.6
fence/hedge	0	857	41	58	2436	307	88	3205	430	70.5	32.8	44.8
roof	90	3439	196	3	50	101942	1627	1165	536	95.9	93.5	94.7
facade	8	693	84	45	25	1760	6758	1509	342	71.1	60.2	65.2
shrub	0	4101	105	105	466	714	339	15107	3881	45.4	60.9	52
tree	5	1401	1	8	66	1023	321	5921	45480	86.8	83.9	85.3

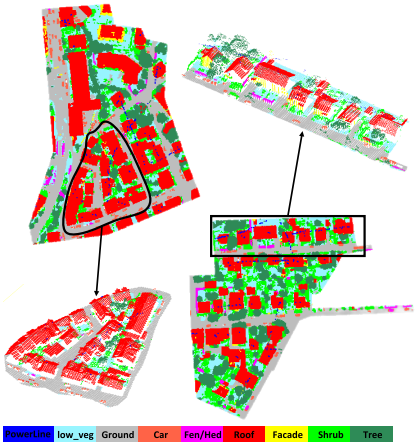


Fig. 7. Classification results of the proposed RRDAN on the ISPRS benchmark dataset. The parts in the black boxes are enlarged to show the details, especially the classification results of facade (yellow), car (tomato), and powerline (blue).

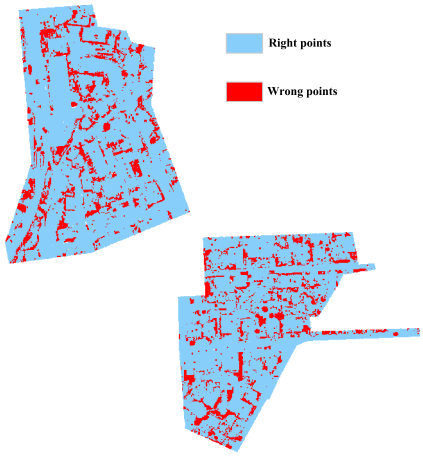


Fig. 8. Error map of the proposed RRDAN on the ISPRS benchmark dataset.

precision, recall, and F1 scores are all above 80%. Noticeably, the categories of fence/hedge, facade, and shrub achieve the worst classification results. According to the confusion matrix, a large proportion of fence/hedge is incorrectly identified as shrub because the two categories have the intricate and alternating relationship, and similar elevation distributions. The category of facade is main mistaken for shrub and roof. The main reason is that facade and shrub are relatively close to each other, while roof and facade have similar physical properties. In addition, the category of shrub is likely to be misidentified as low vegetation and tree since they have similar topographic attributes and reflectance intensities.

To visualize the results of Table II, Figs. 7 and 8 show the classification map and error map, respectively. We can see that most of the points are classified correctly. As shown in the top right of Figs. 7 and 8, the visualized results of low vegetation, impervious surfaces, roof, and tree are also well because the proposed network makes full use of multilevel information to improve the diversity and discrimination of features.

3) Comparisons With Other Methods: To evaluate the performance of RRDAN, we compared with other published methods that achieved the best results at that time, as shown in Table III. These methods mainly have three categories, including PointNet-based methods, KPConv-based methods, and other classic methods. The PointNet-based methods, including DPE, PointCNN, GANet, DANCE-NET, GraNet, RandLANet, VD-LAB, RFFS-Net, and D-FCN, are based on the pointwise MLP structure. The KPConv series of methods include KPConv, LGENet, and our proposed method, which are based on the KPConv framework. In addition, we also compared with some other classic methods, including LUH, WhuY4, RIT_1, and alsNet. The results of the compared methods are obtained from the corresponding papers.

As shown in Table III, for the PointNet-based methods, GANet and GraNet achieve better results than the other PointNet-based methods in Avg.F1. The main reason is that both of the two methods designed an elevation attention module based on the different height information of point clouds to enhance the representation performance of height information. Meanwhile, our proposed RRDAN generates higher Avg.F1 than the compared PointNet-based methods. Because RRDAN not only uses the reflection intensity combined with the elevation information but also aggregates the local accumulation information, which can effectively learn the intrinsic diversified features of point cloud data.

For the KPConv-based methods, LGENet increases the OA and Avg.F1 of 0.1% and 2.4% than the basis KPConv network, respectively. The advantage of LGENet is that the

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TABLE III

COMPARISON RESULTS BETWEEN RRDAN AND THE OTHER METHODS ON THE ISPRS BENCHMARK DATASET. THE RESULTS INCLUDE THE F1 SCORES (%) OF EACH CATEGORY, THE OAS (%), AND THE AVG.F1 SCORES (%). IN THE COLUMN, THE BOLD FONT DENOTES THE HIGHEST RESULTS, AND THE UNDERLINE IS THE SECOND BEST RESULTS

Methods	power	low_veg	imp_surf	car	fence/hedge	roof	facade	shrub	tree	OA	Avg.F1
LUH [5]	59.6	77.5	91.1	73.1	34.0	94.2	56.3	46.6	83.1	81.6	68.4
WhuY4 [15]	42.5	82.7	91.4	74.7	**53.7**	94.3	53.1	47.9	82.8	**84.9**	69.2
RIT_1 [11]	37.5	77.9	91.5	73.4	18.0	94.0	49.3	45.9	82.5	81.6	63.3
alsNet [18]	70.1	80.5	90.2	45.7	7.6	93.1	47.3	34.7	74.5	80.6	60.4
KPConv	68.2	82.2	91.4	72.9	39.9	**94.9**	**65.2**	44.2	82.4	84.4	71.3
DPE [3]	68.1	**86.5**	**99.3**	75.2	19.5	91.1	44.2	39.4	72.6	83.2	66.2
PointCNN [9]	61.5	82.7	91.8	75.8	35.9	92.7	57.8	49.1	78.1	83.3	69.5
GANet [23]	75.4	82.0	91.6	77.8	44.2	94.4	61.5	49.6	82.6	84.5	73.2
DANCE-NET [6]	68.4	81.6	92.8	77.2	38.6	93.9	60.2	47.2	81.4	83.9	71.2
D-FCN [9]	70.4	80.2	91.4	78.1	37.0	93.0	60.5	46.0	79.4	82.2	70.7
RandLANet [21]	68.8	82.1	91.3	76.6	43.8	91.1	61.9	45.2	77.4	82.1	70.9
GraNet [21]	67.7	82.7	91.7	80.9	51.1	94.5	62.0	49.9	82.0	84.5	73.6
VD-LAB [26]	69.3	80.5	90.4	79.4	38.3	89.5	59.7	45.7	77.2	81.4	70.2
RFFS-Net [27]	75.5	80.0	90.5	78.5	45.5	92.7	57.9	48.3	75.7	82.1	71.6
LGENet [13]	**76.5**	82.1	91.8	80.0	40.6	93.8	64.7	49.9	83.6	84.5	73.7
Ours (RRDAN)	72.2	81.7	91.2	**84.6**	44.8	94.7	**65.2**	**52.0**	**85.3**	**84.9**	**74.6**

edge conditional convolution is used to reveal the global–local context relationship and the hybrid point convolution to learn the representative features of point clouds. In addition, LGENet also shows better performance than the compared PointNet-based methods in Avg.F1, which indicates that the KPConv is more suitable for point cloud data than the standard convolution in the PointNet-based methods, where the standard convolution needs to transform the point cloud data into the type of regular data. However, the proposed RRDAN network achieves higher accuracies compared with LGENet; in particular, the F1 scores for the categories of car, fence, shrub, and tree increase by 4.6%, 4.2%, 2.1%, and 1.7%, respectively. This is mainly due to the proposed novel strategy on KPConv through the kernel point attention and the recurrent feature aggregation.

In addition, the proposed RRDAN also shows advantages for point cloud segmentation compared with some other classic methods in Table III. Meanwhile, our designed network generates the best OA and Avg.F1 in all the compared methods; in particular, RRDAN possesses more prominent advantages in the categories of car, roof, facade, shrub, and tree. For the proposed RRDAN, it can make full use of the distribution properties of point clouds to learn the diversified features from the aggregation of local accumulation information and achieve the information interaction of global–local features.

D. Experiment on the DFC2019 Dataset

1) Implementation Details: We further used the DFC2019 dataset to demonstrate the generalization capability of the proposed method. According to [9], we randomly selected 100 from 110 labeled regions on the DFC2019 dataset as the training set and the remaining ten regions as the test set. To achieve robust results, we conducted ten repeated experiments to obtain different training and test sets. For this dataset possessing larger objects and more points, we set the radius of the sphere and the grid interval to 20 and 0.5 m, respectively. In addition, the number of epochs is set to 150 with 800 iterations in each epoch, and we adopted the same manner as ISPRS to train the network.

TABLE IV

COMPARISON RESULTS ON THE DFC 2019 DATASET. THE RESULTS INCLUDE THE F1 SCORES (%) OF EACH CATEGORY, THE OAS (%), AND THE AVG.F1 SCORES (%). IN THE COLUMN, THE BOLD FONT DENOTES THE HIGHEST RESULTS, AND THE UNDERLINE IS THE SECOND BEST RESULTS

Methods	Ground	Hig_veg	Building	Water	Bridge	OA	Avg.F1
PointNet++ [9]	98.3	95.8	79.7	4.4	7.3	92.7	57.1
PointCNN [6]	98.7	97.2	84.9	44.1	65.3	93.8	78.0
PointConv [26]	97.3	95.8	93.6	74.5	69.2	95.3	86.1
KPConv	98.4	94.2	87.4	43.0	77.5	94.5	80.1
D-FCN [9]	99.1	98.1	89.9	45.0	73.0	95.6	81.0
DANCE-NET [6]	99.1	93.9	87.0	58.3	**83.9**	96.8	84.4
DGCNN [27]	97.9	93.2	90.4	**88.2**	54.4	95.1	84.2
LGENet [13]	**99.3**	**98.3**	92.8	47.4	79.1	**98.4**	83.4
LGGAA [47]	98.9	96.1	90.2	41.6	83.7	94.8	81.4
DA-Net [8]	99.3	97.6	92.7	41.6	85.1	98.3	83.3
ours (RRDAN)	99.1	98.1	**95.8**	62.8	82.3	98.1	**87.6**

2) Classification Results: To demonstrate the effectiveness of the proposed model, we compared with PointNet++, PointSIFT, PointCNN, KPConv, D-FCN, DANCE-NET, DGCNN, LGENet, LGGAA, and DA-Net. The segmentation results are given in Table IV and Fig. 9, and Table V shows the results of ten repeated experiments.

From Table IV, the proposed network achieves the best performance in Avg.F1 than the other compared methods and the second best in OA. Specifically, compared with the basis

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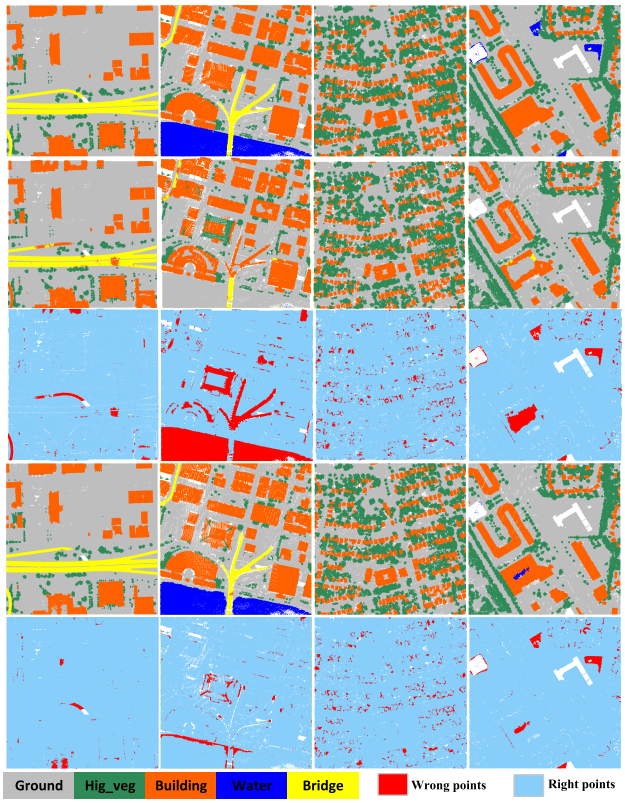


Fig. 9. Classification results for several regions on the DFC2019 dataset. The first, second, third, fourth, and fifth rows are ground truth, classification map for KPConv, error map for KPConv, classification map for RRDAN, and error map for RRDAN, respectively.

TABLE V

RESULTS OF TEN REPEATED EXPERIMENTS ON THE DFC2019 DATASET. THE RESULTS INCLUDE THE F1 SCORES (%) OF EACH CATEGORY, THE OAS (%), AND THE AVG.F1 SCORES (%)

No.	Ground	Hig_veg	Building	Water	Bridge	OA	Avg.F1
1	99.4	98.4	96.2	24.9	90.6	98.5	81.9
2	98.4	97.5	95.8	72.4	87.3	97.0	90.3
3	99.5	98.8	95.8	94.4	94.3	98.8	96.5
4	99.5	97.5	95.9	78.5	96.7	98.5	93.6
5	99.3	97.4	93.7	92.8	85.9	97.6	93.8
6	99.4	97.5	96.8	46.5	55.2	98.6	79.1
7	97.8	98.6	96.2	34.5	73.1	96.5	80.1
8	99.5	98.2	96.1	76.2	91.5	98.5	92.3
9	98.7	98.3	94.6	85.6	72.1	97.6	89.8
10	99.8	98.3	96.4	21.8	76.3	98.9	78.5
Average	99.1	98.1	95.8	62.8	82.3	98.1	87.6

KPConv, the F1 scores of the proposed RRDAN for ground, high-vegetation, building, water, and ridge-deck increase by 0.7%, 3.9%, 8.4%, 19.8%, and 4.5%, respectively. For the proposed network, its OA is the third best than the other networks, which is just lower 0.3% than the best result (i.e., LGENet), while its Avg.F1 exceeds 1.5% and 3.2% compared with two best comparison networks (i.e., PointConv and DANCE-NET, respectively).

In Fig. 9, the wrong points are very small and the classification results are closer to their ground truths. These results demonstrate the effectiveness of the proposed RRDAN model for the large-scale ALS point cloud data. In Table V, the fluctuation of the repeated experimental F1 scores is very small for ground, high-vegetation, and building, but large for water and bridge-deck, because the two categories have fewer points than the other categories resulting in the uneven distribution of samples for model training.

TABLE VI

COMPARISON RESULTS ON THE LASDU DATASET. THE RESULTS INCLUDE THE F1 SCORES (%) OF EACH CATEGORY, THE OAS (%), AND THE AVG.F1 SCORES (%). IN THE COLUMN, THE BOLD FONT DENOTES THE HIGHEST RESULTS, AND THE UNDERLINE IS THE SECOND BEST RESULTS

Methods	Ground	Building	Tree	Low_veg	Artifact	OA	Avg.F1
PointNet++[44]	87.7	90.6	82.0	63.2	31.3	82.8	71.0
PointCNN[26]	89.3	92.8	84.1	62.8	31.7	85.0	72.1
PointConv[26]	89.6	94.3	84.6	67.5	36.4	85.9	74.5
DGCNN[27]	90.5	93.2	81.6	63.3	37.1	85.5	73.1
ShellNet[26]	89.5	92.8	84.6	**68.5**	35.7	85.6	74.2
PosPool[26]	88.3	93.7	83.9	61.0	38.3	83.5	73.0
KPConv	89.9	96.1	**85.1**	56.9	33.6	85.9	72.3
ours (RRDAN)	**91.6**	**96.6**	84.1	66.3	**48.3**	**87.7**	**77.4**

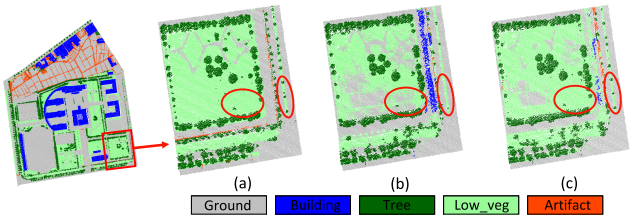


Fig. 10. Classification maps of a local enlarged region under different models on partial LASDU dataset. (a) Ground truth. (b) KPConv. (c) Ours.

E. Experiment on the LASDU Dataset

To demonstrate the generalization of the proposed model, we compared it with previous advanced models using the LASDU dataset. Note that the results of the methods in Table VI were obtained from the corresponding papers. Fig. 10 shows the effect of our network compared with the KPConv. From Table VI, the proposed network achieves the best performance in OA and Avg.F1 than the other compared methods. Specifically, compared with the basis KPConv, the F1 scores of the proposed RRDAN for ground, building, low_veg, and artifact increase by 1.7%, 0.5%, 9.4%, and 14.7%, respectively. For the proposed network, its Avg.F1 exceeds 2.9% and 3.2% compared with two best comparison networks (i.e., PointConv and ShellNet, respectively). In Fig. 10, the classification results of enlarged regions are closer to their ground truth. These results demonstrate the effectiveness of RRDAN for large-scale ALS point cloud data.

F. Ablation Study

In this section, on the ISPRS dataset, we investigated the effectiveness of the proposed three modules, i.e., AKPConv, RRKA, and GLCA, where RRKA is constructed based on the proposed AKPConv block and the recurrent residual operation.

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TABLE VII
ABLATION ANALYSIS FOR DIFFERENT MODULES ON THE ISPRS DATASET

Model	KPConv	AKPConv	RRKA	GLCA	γ	Focal	CE	OA(%)	Avg.F1(%)	Params	FLOPs	Training time
A		√	√	√	√	√		84.9	74.6	74.92M	13.16G	4.18h
B		√	√	√		√		84.3	73.1	74.92M	13.16G	4.18h
C	√		√	√	√	√		84.3	73.9	74.91M	13.16G	3.99h
D		√		√	√	√		84.4	74.1	24.41M	6.68G	2.32h
E		√	√		√	√		84.6	71.6	74.92M	13.16G	3.47h
F	√						√	84.4	71.3	24.40M	6.68G	2.52h
G	√				√		√	84.4	71.6	24.40M	6.68G	2.52h
H		√	√	√	√		√	84.9	73.6	74.92M	13.16G	4.18h

We also analyze the complexity of each model, including the number of model parameters (Params), training time, and floating point of operations (FLOPs). Table VII shows the segmentation results under different modules. Model A is the proposed RRDAN, model B is the case of RRDAN without gamma transformation ($\gamma = 1$), model C denotes the AKPConv of RRDAN replaced by KPConv, model D is the RRDAN without RRKA operation, model E is the RRDAN without GLCA, model F is the KPConv network, model G is the model F with gamma transformation ($\gamma = 0.5$), and model H replaces the focal loss of model A with the cross-entropy (CE) loss.

In Table VII, model A has higher OA and Avg.F1 over 0.6% and 1.5% than model B, respectively, and model G is over 0.3% in Avg.F1 than model F, which indicates that the gamma transformation can improve the distribution to enhance the network training. The OA and Avg.F1 of model C are lower than those of model A, which indicates that the proposed AKPConv with attention is benefit to highlight the contribution of key features. For model D, it removes the RRKA operation in RRDAN to achieve lower OA and Avg.F1 than model A because the RRKA operation can aggregate the local information to enhance the diversity of features. Comparing model A with model E, model A is superior to model E in terms of OA and Avg.F1. The reason is that the proposed GLCA module can perform the interaction of global and local information to improve the fusion between the low- and high-level features. To further explain the effectiveness of the proposed modules, we also utilize the previous KPConv architecture as the base network. Model A (i.e., RRDAN) improves 3.3% in Avg.F1 and increases OA from 84.4% to 84.9% compared with model F (the base network), which reveals that our proposed network can effectively enhance the local and global feature representation of point cloud data to promote the discrimination performance of features. Comparing model A with model H, model A has better Avg.F1 to verify that the focal loss is suitable for the unbalanced sample distribution of point clouds.

In addition, we also analyzed the impact of model parameter size, training time, and computational efficiency. The results show that model D with AKPConv only increased parameter size by 0.01M compared to the baseline model F with KPConv while achieving a 2.8% improvement in Avg.F1 precision. For model A, its parameter size, training time, and FLOPs increase to 74.92M, 4.18 h, and 13.16G, respectively, which indicates the RRKA module with high complexity. However, model A achieves best OA and Avg.F1 than the other models. For ALS point clouds, the complexity is acceptable to the accuracy improvement under sufficient computational resources.

TABLE VIII
RESULTS OF RRDAN WITH DIFFERENT NUMBERS OF RECURRENCE ON THE ISPRS DATASET

parameter	power	low_veg	imp_surf	car	fence/hedge	roof	facade	shrub	tree	OA	Avg.F1
$t = 3$	70.8	**81.7**	**91.3**	68.1	42.8	93.9	64.5	48.9	82.1	83.9	71.6
$t = 4$	**72.2**	**81.7**	91.2	**84.6**	**44.8**	**94.7**	**65.2**	**52.0**	**85.3**	**84.9**	**74.6**
$t = 5$	68.5	**81.7**	91.1	75.7	39.9	94.6	63.5	50.6	83.6	84.2	72.1

According to Table VII, Fig. 11 shows the classification maps of a local enlarged region under different models on the ISPRS dataset. In this figure, the completed model A shows the best classification result than the other several models and has the closest visualized result to ground truth. In particular, in the circle area, the main two categories are car and facade, model A has fewer misclassified points than the other models, which indicates the effectiveness of each proposed module.

G. Parameter Analysis

The proposed method mainly has two parameters, including the number of recurrences t in the RRKA module and the gamma transformation parameter γ. We analyze their influence on the results of the ISPRS dataset.

1) Number of Recurrence: For the number of recurrences, we gave the results of RRDAN with $t = 3, 4, 5$ in Table VIII, and a local classification result is shown in Fig. 12.

According to Table VIII, with the increased t, the OA and Avg.F1 first improve and then decline. The reason is that the network can aggregate more information with the increasing of t, while an excessive t will result in the worthless information obtained by the network to reduce the discriminating power of features. Therefore, it is very important to set a proper number of recurrence. In Table VIII, when t is 4, it shows the best OA and Avg.F1 and has a better F1 score under most of the categories. According to Fig. 12, we can also achieve the closest classification map to ground truth with $t = 4$. Thus, we set the number of recurrences to 4 in this article.

2) Gamma Transformation Analysis: In the gamma transformation, there is a significant parameter γ to influence the correction precision of reflection intensity. A large or small γ causes the long-tailed feature distribution. In this section,

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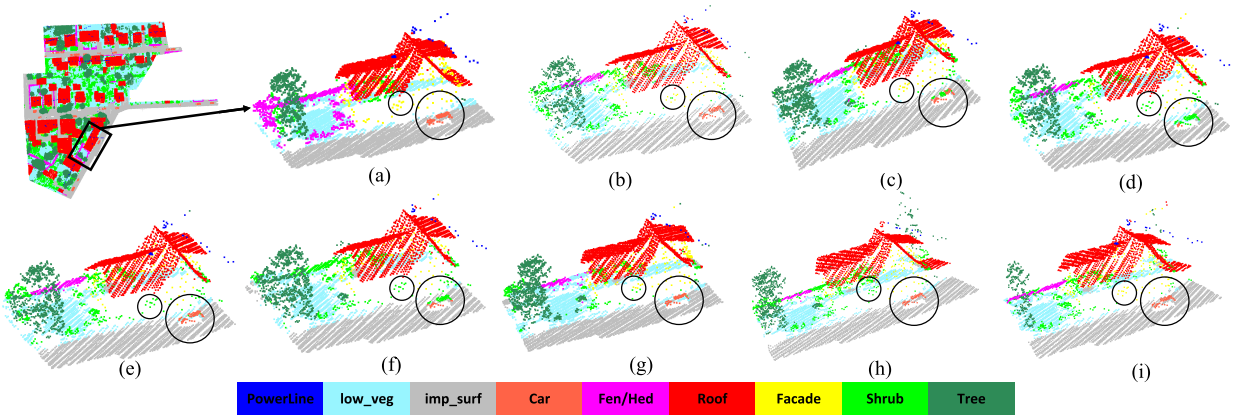


Fig. 11. Classification maps of a local enlarged region under different models on the ISPRS dataset. (a) Ground truth. (b) Model A. (c) Model B. (d) Model C. (e) Model D. (f) Model E. (g) Model F. (h) Model G. (i) Model H.

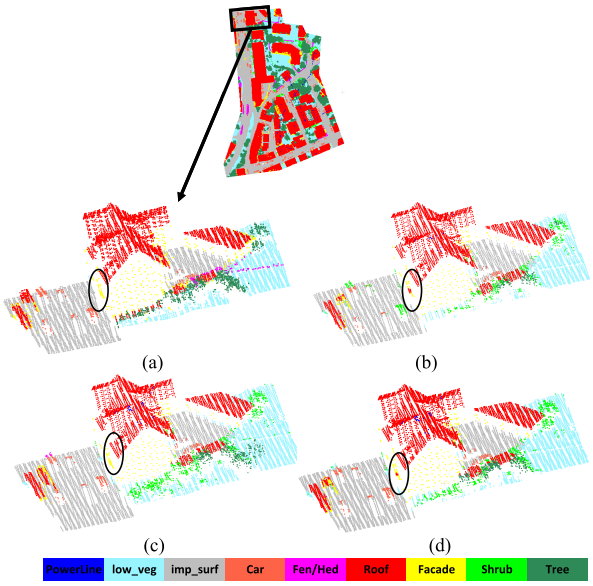


Fig. 12. Classification results of a local enlarged region with different numbers of recurrence on the ISPRS dataset. (a) Ground truth. (b) $t = 3$. (c) $t = 4$. (d) $t = 5$.

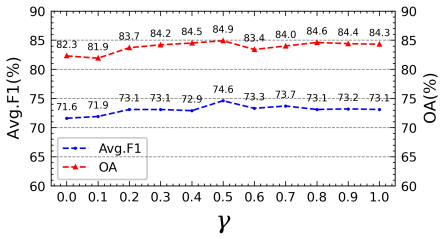


Fig. 13. Results for different gamma of the ISPRS benchmark dataset.

we conducted some experiments to analyze the classification results under different γ, where γ is set to 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, and 1. When γ is set to 1, it denotes to perform the experiment without gamma transformation. The results are shown in Fig. 13.

According to Fig. 13, the change of OA and Avg.F1 is first increasing and then decreasing approximately. When γ is set to 0.5, the OA and Avg.F1 both have the best result than

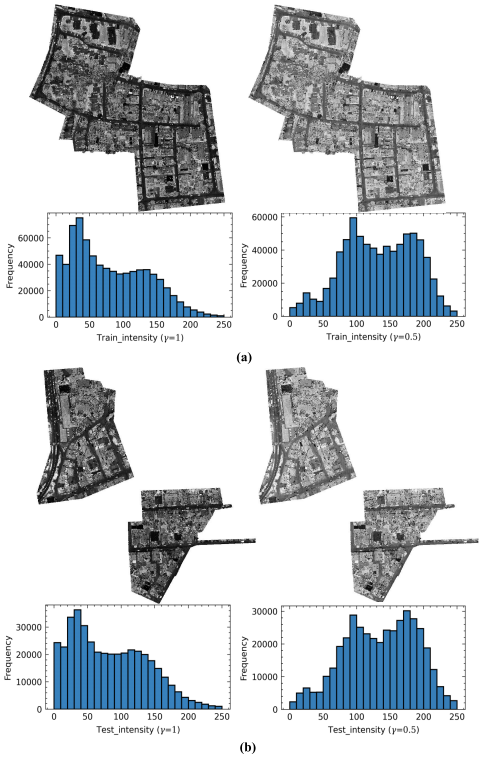


Fig. 14. (a) Original training set distribution on the left and gamma transformed training set distribution on the right. (b) Original test set distribution on the left and gamma transformed test set distribution on the right.

the other values. Because an improper γ will make the over concentration of data distribution to reduce the diversity of features. Therefore, γ should be set to a proper value, and in this article, we use γ as 0.5. In Fig. 14, it shows the original distribution ($\gamma = 1$) and the gamma transformed distribution ($\gamma = 0.5$) of training and test sets. In this figure, the original data present the long-tailed feature distribution, which will result in a lack of diversity for network feature learning. After preprocessing by gamma transformation, the intensity features meet the normal distribution to reduce the preference of network and improve the diversity and discrimination of features. Therefore, we can obtain better OA and Avg.F1 with the gamma transformed intensity than the original intensity, as shown in Fig. 13.

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IV. Conclusion

In this article, we propose an RRDAN based on the encoder–decoder structure. According to KPConv, we construct an AKPConv module based on the scale factor of batch normalization to highlight the key features of points. Then, based on AKPConv, an RRKA module is designed with the recurrent operation to aggregate the local information and enhance the diversity of features. To better fuse the low- and high-level sematic features, we develop a GLCA to achieve the information interaction and enhance the discrimination power of features. According to AKPConv, RRKA, and GLCA, a cloud point sematic segmentation network (i.e., RRDAN) is proposed with the encoder–decoder structure. In addition, we adopt gamma transformation to preprocess the reflection intensity features, which will correct the intensity feature distribution to improve the fairness of network learning. The proposed RRDAN not only learns the diversitied features but also performs the interaction of global and local features to improve the sematic segmentation performance of ALS point cloud data. To demonstrate the benefits of the proposed network, we conducted experiments on the ISPRS, DFC2019, and LASDU datasets. The experimental results show that the proposed RRDAN achieves better segmentation performance in OA and Avg.F1 compared to a few advanced approaches. Although RRDAN has a powerful segmentation capability for ALS point clouds, it has a larger number of network parameters. In the future, we would like to design a lightweight RRDAN model, i.e., we can introduce the knowledge distillation [48] technique into the RRDAN to reduce the number of network parameters.

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